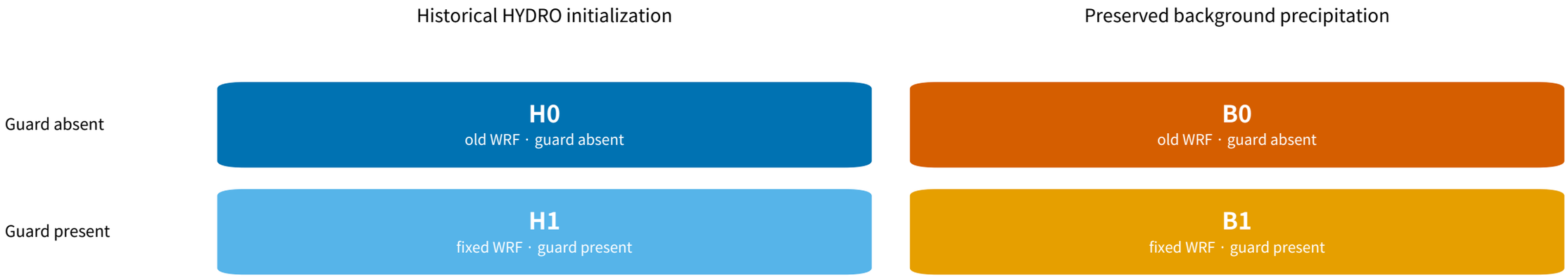
Experimental structure and metric definition

Factorial cells: initialization policy × zero-rate guard



O0 · BASE_ORIGINAL: actual original OFF forecast (external anchor)

What changes in the intervention

Preserved-background initialization changes eight coupled native fields: P, PH, AL, P_HYD, TH2, QRAIN, QSNOW, QGRAUP.

Separate QR/QS swaps are prescribed species diagnostics; they do not represent the full policy intervention.

Zero-rate fix: fallback runs only when the rate $r \neq 0$; exact $+0$ and -0 bypass it (no maximum clamp).

Contrasts: guard H1−H0 / B1−B0; policy B0−H0 / B1−H1. Interaction is a paired score difference-in-differences, not a physical causal share.

Rainfall and verification quantities

$$P_{12\to13} = (RAINC + RAINNC)_{13UTC} - (RAINC + RAINNC)_{12UTC}$$

Cumulative WRF totals are differenced after reset checks; RAINSH is checked zero and excluded; tolerance-level negatives are set to zero.

$$CSI(\tau) = H/(H + M + F), \quad H = \#(P \geq \tau,\ R \geq \tau)$$

Each model total P and processed RN1 value R is thresholded at the same 1-h accumulation τ (mm); no interpolation or coverage extrapolation.